
Acoustic Anomaly Detection and Fault Classification in Industrial Machinery

A
Dissertation Report
Submitted in Partial Fulfillment of the Requirements
for the Degree
of
MASTER OF SCIENCE IN DATA SCIENCE
FROM
POSTGRADUATE DEPARTMENT OF DATA SCIENCE
ST. XAVIER'S COLLEGE (AUTONOMOUS), KOLKATA

by
DEBJIT KHASKEL
CIN: 24-500-5-08-0438
REG NO: A01-1112-0152-24

UNDER THE SUPERVISION OF

DR. MD. SAHIDULLAH
ASSISTANT PROFESSOR
INSTITUTE FOR ADVANCING INTELLIGENCE
(IAI), TCG CREST, KOLKATA

INTERNAL MENTOR

DR. ROMIT S BEED
ASSOCIATE PROFESSOR
ST. XAVIER'S COLLEGE (AU-
TONOMOUS), KOLKATA



DECLARATION

I certify that

- a. The work contained in the thesis is original and has been done by myself under the general supervision of my supervisor.
- b. the work has not been submitted to any other institute for any degree or diploma.
- c. I have followed the guidelines provided by the institute in writing the report.
- d. I have conformed to the norms and guidelines given in the ethical code of conduct of the institute.
- e. Whenever I have used materials (data, theoretical analysis, text and images) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references.
- f. Whenever I have quoted written materials from other sources, I have put them under quotation marks and given due credit to the sources by citing them and giving the required details in the references.
- g. I used AI tools for proofreading and formatting my report, ensuring perfect alignment and presentation.

Date:

Place:

Debjit Khaskel
CIN: 24-500-5-08-0438
Reg No: A01-1112-0152-24

Acknowledgements

I sincerely express my gratitude to my supervisor, **Dr. Md. Sahidullah**, from TCG Crest, Kolkata, for his guidance, insightful feedback, and constant support throughout my internship and dissertation period. Under his mentorship, I have explored different aspects of learning and gained a deeper understanding of my field. This journey has been an incredible opportunity for growth, and I am truly grateful for his encouragement and expertise.

I am also deeply thankful to my internal supervisor, **Dr. Romit S Beed**, whose teaching from day one made this course highly engaging and intellectually stimulating. Her continuous support and cooperative nature have made this academic endeavor both smooth and enjoyable.

A heartfelt thanks to **my parents** for their unconditional love, encouragement, and unwavering support, which have been my greatest sources of strength.

I also extend my gratitude to all **my teachers, friends, and classmates**, whose valuable discussions, motivation, and camaraderie have enriched my academic journey. Their presence has made this experience truly memorable.

Finally, I acknowledge everyone who contributed directly or indirectly to the successful completion of my dissertation.

Date:

Place:

Debjit Khaskel
CIN: 24-500-5-08-0438
Reg No: A01-1112-0152-24

Abstract

This work addresses the problem of monitoring industrial machinery through acoustic signals for predictive maintenance, early fault detection, and operational safety. Such an approach is applicable across manufacturing plants, process industries, and automated systems where continuous manual inspection is costly and inefficient. By leveraging machine sound, the system enables continuous, non-intrusive monitoring and supports timely maintenance decisions. The study proposes a learning-based framework for distinguishing normal and abnormal machine behaviour while also providing deeper insight into abnormal operating patterns. The approach first learns informative representations from audio signals and then performs anomaly detection to separate normal and faulty conditions. In addition, the detected anomalies are further analyzed to uncover underlying structures and variations, enabling a more informative understanding of machine faults beyond simple detection. Experimental results demonstrate a substantial improvement in detection performance, achieving approximately 28% relative gain over baseline approaches in terms of overall anomaly detection accuracy. The method also shows improved consistency under varying operating conditions and provides more interpretable insights into fault patterns. In conclusion, the proposed framework offers a practical and scalable solution for intelligent acoustic monitoring of industrial systems, supporting more reliable fault detection and more informed maintenance strategies.

Keywords— Anomaly Detection, Unsupervised Learning

Table of Contents

Declaration	i
Acknowledgements	iii
Abstract	v
Table of Contents	vi
List of Tables	x
Chapter 1 Introduction	2
1.1 Overview	2
1.2 Thesis Structure	3
1.3 Motivation	4
1.4 Scope of work	5
1.5 Contributions	6
Chapter 2 Literature Review	8
2.1 Unsupervised Anomaly Detection for Industrial Audio	9
2.2 Dataset and Domain Shift	10
2.3 Clustering-Based Anomaly Detection	12
Chapter 3 Baseline Anomalous Sound Detection System	13

3.1	Introduction	13
3.2	Dataset: MIMII Pump	14
3.3	Feature Extraction	15
3.3.1	Multi-Channel Averaging and Resampling	15
3.3.2	Log-Mel Spectrogram Computation	16
3.3.3	Temporal Windowing	17
3.3.4	Normalisation	18
3.4	Baseline Model Architecture	19
3.4.1	Encoder	19
3.4.2	Decoder	20
3.4.3	Architecture Summary	21
3.5	Training Procedure	21
3.6	Anomaly Scoring	22
3.7	Evaluation Protocol and Metrics	23
3.8	Experimental Results	24
3.8.1	Performance Comparison Across Machine IDs	24
3.9	Analysis and Discussion	26
Chapter 4	Proposed System: AST with Complete Deep SVDD	28
4.1	Introduction and Motivation	28
4.2	System Overview	29
4.3	Audio Spectrogram Transformer (AST)	30
4.3.1	Architecture and Pretraining	30
4.3.2	Embedding Dimensionality	31
4.3.3	Domain Adaptation via Masked Spectrogram Prediction	32
4.4	Embedding Extraction and Preprocessing	34

4.5	Complete Deep Support Vector Data De- scription	35
4.5.1	Motivation and Theoretical Background	35
4.5.2	Network Architecture	36
4.5.3	Three-Stage Training Protocol	37
4.6	Evaluation Protocol	39
4.7	Experimental Results	40
4.7.1	Proposed System Performance Across Machine IDs	40
4.7.2	Comparison with the Baseline System	41
4.8	Discussion	42
Chapter 5	Post-Hoc Fault Subtype Discovery via Clus- tering	44
5.1	Introduction and Motivation	44
5.2	Data and Embedding Spaces	45
5.2.1	Anomalous Sample Set	45
5.2.2	AST Embedding Space	46
5.2.3	CAE Embedding Space	46
5.2.4	Dimensionality Reduction Pipeline	47
5.3	Primary Method: HDBSCAN Clustering	48
5.3.1	Algorithm and Parameter Choice	48
5.3.2	HDBSCAN Results for pump id_00	48
5.3.3	Spectral Characterisation of AST Clus- ters (id_00)	50
5.4	Secondary Method: Deep Embedded Clus- tering with Generalizability-Based k Selection	51
5.4.1	Motivation for DEC	51
5.4.2	Generalizability-Based k Selection	52
5.4.3	DEC Architecture and Training	52
5.4.4	DEC Results for pump id_00	54

5.5	Method Agreement Analysis	55
5.5.1	HDBSCAN vs DEC Agreement on AST Embeddings	55
5.5.2	Cross-Embedding Agreement	56
5.6	Results Across All Machine IDs	57
5.6.1	Consolidated HDBSCAN Summary	57
5.6.2	CAE Clustering Across Machine IDs	58
5.6.3	Per-Machine DEC Validity and Agree- ment	59
5.6.4	Spectral Characterisation Across Ma- chine IDs	60
5.7	Discussion	61
Chapter 6	Conclusion and Future Work	62
6.1	Conclusion	62
6.2	Future Work	63

List of Tables

3.1	Dataset split for MIMII pump id_00 (0 dB SNR). The model is trained exclusively on normal files; abnormal files appear only at evaluation time.	15
3.2	Log-mel spectrogram extraction hyperparameters. . . .	17
3.3	Layer-wise architecture of the convolutional autoencoder baseline.	21
3.4	Baseline convolutional autoencoder vs. MIMII paper (pump, 0 dB SNR).	25
4.1	AST fine-tuning configuration for the masked spectrogram prediction stage.	33
4.2	Architecture of the Deep SVDD network ϕ . All linear layers omit bias terms to prevent hypersphere collapse.	37
4.3	Complete Deep SVDD hyperparameters.	39
4.4	Performance metrics across pump machine IDs.	41
4.5	System comparison on MIMII pump (0 dB SNR). Gain reported relative to the MIMII baseline.	41
5.1	HDBSCAN results on AST and CAE embeddings for pump id_00 (143 anomalous samples). Cluster stabilities are HDBSCAN's internal persistence scores; higher is more stable.	48

5.2	Spectral descriptor statistics per HDBSCAN cluster on AST embeddings, pump id_00. Computed over all non-noise anomalous samples assigned to each cluster. . . .	50
5.3	DEC architecture and training configuration.	53
5.4	Internal clustering validity scores for DEC and HDBSCAN across both embedding spaces, pump id_00. . .	54
5.5	HDBSCAN vs DEC agreement metrics per embedding space, pump id_00. ARI and NMI computed over non-noise HDBSCAN samples only.	55
5.6	HDBSCAN clustering results on AST embeddings across all pump machine IDs (0 dB SNR). k^* (DEC) is the generalizability-selected cluster count. ARI is the HDBSCAN vs DEC agreement on AST, computed over non-noise samples.	57
5.7	HDBSCAN results on CAE embeddings across all pump machine IDs. A noise fraction of 1.000 indicates that HDBSCAN found no density-connected clusters whatsoever.	59
5.8	DEC internal validity scores and HDBSCAN vs DEC agreement across all machine IDs. For id_04 and id_06, CAE ARI/NMI values of 1.0000 are trivially vacuous: both methods found zero CAE clusters (all samples assigned as noise), so there is no partition to compare. .	59
5.9	Per-cluster spectral centroid (Hz, mean $\pm$ std) from HDBSCAN on AST embeddings for id_02 and id_06. The consistent ordering across clusters within each machine indicates spectral separation driven by shape rather than amplitude differences (RMS is identical at 0.01 across all clusters and machines).	60

Introduction

1.1 Overview

Industrial machinery monitoring relies on acoustic data because sound captures fine and minute mechanical interactions. Therefore, the sound or audio produced when a machine works normally is quite different from the sound produced when there is an anomaly in the machine which is hard to detect. After an anomaly is detected, it now becomes the job of the engineer to inspect the totality of the system, pinpoint where exactly the problem has occurred, and determine the type of anomaly the machine has faced, which takes a lot of the time and cost of the industry to fix the anomaly. So we create an autonomous audio-driven diagnostic system that analyzes machine sounds over time [1].

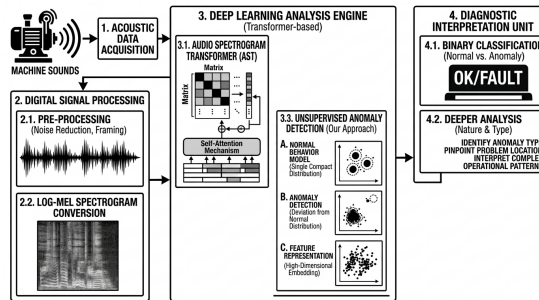


Figure 1.1: Technical workflow of the unsupervised machine sound monitoring and diagnostic pipeline

The MIMII dataset has become a benchmark [2, 3] that leads to the research and development of anomaly detecting techniques using machine acoustic data.

Traditionally, there are two ways to deal with anomaly detection [4, 5], one is the supervised approach, and the other is the unsupervised approach. When we are doing supervised learning we need labeled anomalous data which is rare and scarce in real world environments. Taking this into account, we make use of unsupervised learning methods that learn how a normal machine sounds while operating and flag any deviations from it as anomalous.

With recent advancements in deep learning, especially in transformer-based architectures [6], there has been a significant improvement in feature representation from audio signals. However, traditionally it is assumed that the normal machine audio follows a single compact distribution, thereby creating a discrepancy when dealing with real life scenarios, as any industry has complex operational patterns mixed with various environmental noise. Usually the primary focus is on anomaly detection with going deeper into further analysis of the anomalies which provide deeper insights into its structure or types.

1.2 Thesis Structure

- Chapter 2 (Literature Review): Reviews existing approaches in industrial acoustic anomaly detection, including unsupervised

methods, dataset challenges, and clustering-based analysis techniques.

- Chapter 3 (Baseline System): Presents a baseline anomaly detection system and evaluates its performance, highlighting key limitations of conventional approaches.
- Chapter 4 (Proposed System): Introduces the proposed framework and demonstrates its improvements in anomaly detection performance.
- Chapter 5 (Fault Analysis via Clustering): Explores how detected anomalies can be further analyzed to identify underlying fault patterns and structures.
- Chapter 6 (Conclusion and Future Work): Summarizes the findings and outlines potential directions for extending this work.

1.3 Motivation

In real life scenarios, machine behavior is usually influenced by many factors, some are environmental in nature, some are influenced by various conditions in which they operate and under the varying loads they operate based on the supply and demand of the products in the market for which the machines are used. Therefore, trying to model such data using conventional methods often leads to suboptimal performance and inefficient management of the machines.

In addition, we evaluate anomaly detection systems based on how it is able to distinguish normal sounds from anomaly sounds. Here we see that using only binary classification is not sufficient in real life scenarios, we need a good understanding of the nature and the type of the anomalies we are facing to further help us in diagnosis and maintenance.

Another limitation is that we lack detection systems that stop after the anomalies have been detected and there are no attempts to further investigate them to interpret where and how the anomaly occurred, which results in delayed actions to remedy them even after proper detection, since we lack insight about the anomalies.

Keeping these limitations in mind, we need a deep learning approach which not only helps us extract and understand complex patterns, but also a modeling framework which takes into account how a normal machine behaves in varying environmental conditions. Then we also need to further analyze the anomalies we detected to understand the underlying structure and locate it.

This work focuses on developing a two-stage unsupervised anomaly detection framework for industrial audio data.

1.4 Scope of work

This work focuses on developing an intelligent system that can monitor machine health using sound. The primary objective is to detect

when a machine is operating abnormally and to provide deeper insight into the nature of these abnormalities.

The system first learns how a machine behaves under normal conditions using acoustic data. It then identifies unusual patterns that deviate from this normal behavior. Beyond detection, the work also aims to analyze these abnormal patterns to understand whether different types of faults or conditions can be distinguished.

The key objectives of this work are:

- Detect anomalies in machine audio under varying operating conditions.
- Create an improved baseline system and further improve it to make a better robust system.
- Analyze abnormal sounds to gain insight into different types of faults.

1.5 Contributions

- **Combination of Transformer-based Representation and Deep Classifier:** Combined Audio Spectrogram Transformer (AST) embeddings with a Deep SVDD model to effectively capture complex temporal-frequency patterns and model multi-modal abnormal behavior in industrial audio data.
- **Clustering-driven Anomaly Structure Discovery with**

Empirical Validation: Clustering techniques are applied to detected anomalies to uncover latent subgroups, improving interpretability and potential fault categorization, with validation conducted on the MIMII dataset.

- **Post-Detection Deep Analysis of Anomalous Representations:** Performed an in-depth analysis of the learned embedding space to investigate the structure and distribution of anomalous samples. This analysis provides deeper insight into the characteristics of the anomaly, validates the effectiveness of the learned feature space, and supports the identification of distinct fault patterns.

Literature Review

Industrial machinery is the core of any industry that involves manufacturing, energy production, and process automation. When a failure occurs on fans, pumps, sliders, or rails, it leads to great downtime and incurs significant cost. Therefore, the use of a system that continuously monitors the machines has become essential so that the faults do not accumulate and become too big to handle when left undetected.

Traditionally, to monitor machine condition, vibration analysis was performed, as it was used to detect various types of fault, mainly bearing related or shaft misalignment. But installing a vibration sensor for each of the machines is quite expensive and sometimes it becomes impractical, the vibration sensors also activate when a fault has already occurred, it cannot detect before a fault occurs.

With the emergence of Acoustic monitoring we found an effective way to perform the task in a non-contact format with ease of deployment. We make use of microphones that can capture all sorts of audio data, including machine vibrations, rotatory noise, and other machine audio. Even with these advantages, it becomes quite challenging to perform this acoustic monitoring, as the audio is usually

mixed with environmental noise, and sometimes even speech data are mixed within it.

In this section, we will explore previous research done on unsupervised anomaly detection [7, 5, 8] for industrial audio, the industrial audio data collection and domain shift, and clustering based fault detection, diving deeper into this method, and discussing its key aspects.

2.1 Unsupervised Anomaly Detection for Industrial Audio

In the field of predictive maintenance, acoustic monitoring has gained a very high importance. Machinery acoustic data have many and various kinds of information on behavior of the machines, and its abnormal acoustic patterns which tell us about the faults such as contamination, clogging, leaking, or any kind of damage. most of the research on industrial acoustic analysis are unsupervised or semi-supervised in nature, as usually the anomalous acoustic data are rare and hard to localize or collect, so models are trained on normal machinery acoustic data.

In the paper by Koizumi et al. [Koizumi'2019] we see an early framework for unsupervised anomalous sound detection, where machine sounds are converted to spectrograms and fed to a deep learning model which learns the normal operating audio of the machine, and

any point varying from these normal audio is flagged as anomalous. As a result, this established a benchmark in the fields of anomalous sound detection in the Detection and Classification of Acoustic Scenes and Events (DCASE) challenge [Koizumi'2019, 3] and further led to many robust and rigorous studies related to industrial machinery acoustic analysis.

The baseline system introduced in the DCASE 2020 Challenge Task 2 establishes a fundamental reference point for anomalous sound detection using an autoencoder-based approach [9, 10]. This system, evaluated on industrial machine datasets including MIMII, achieves an average performance of approximately 70.5% AUC, highlighting moderate detection capability under controlled conditions.

These results indicate that while reconstruction-based methods can capture general acoustic patterns of normal machine operation, their ability to robustly distinguish anomalies remains limited, particularly in complex or variable environments. This limitation has motivated subsequent research toward more expressive representation learning and improved anomaly modeling techniques.

2.2 Dataset and Domain Shift

In order to do acoustic anomaly detection we need a proper dataset which fulfills all the requirements for doing such task, we make use of the MIMII (Malfunctioning Industrial Machine Investigation and

Inspection) dataset created by Purohit et al. [2] In this dataset we find both normal and anomalous acoustic recordings of many types of industrial machine including pumps, fans, valves, and slide rails. These acoustic data were recorded in real industrial environments with background noise, which provides realistic conditions for anomaly detection.

The MIMII dataset is the most widely used dataset when doing research on anomalous sound detection. It helps in developing many machine learning models that are trained on normal acoustic data, which are then evaluated. This also enables in detecting mechanical faults.

The original MIMII dataset is a good benchmark for anomaly detection, but when faced with real life situations involving industrial environmental noise and conditions, we make use of the MIMII-DUE dataset [11], which was introduced to analyze anomalous acoustic data under domain shift conditions. This dataset acts as an add on to the original MIMII dataset as it includes various domains and conditions which the machines have to face throughout the year.

We usually observe domain shifts when the properties of the training and testing data vary statistically as a result of the changes in background noise, how the audio is being recorded, or machine operating states. this significantly affects the performance of anomaly detection systems. So we use the MIMII-DUE dataset [11] which provides a more realistic approach for evaluating and making acoustic

anomaly detection models.

2.3 Clustering-Based Anomaly Detection

In addition to anomaly audio detection, our next focus is on getting an idea of anomalies using clustering methods [12, 13].

In the paper by Vanem [12] we see that an unsupervised anomaly detection approach using clustering techniques was proposed [12], where we see that first all the acoustic data representing the normal functioning of the machine were clustered, and then the acoustic data points that do not fall under the clusters are treated as anomalies. Since we are performing unsupervised learning the clustering technique [14, 13] helps in finding hidden structures in complex real life datasets and is useful for extracting previously unknown patterns.

For anomaly pattern analysis, we generally use clustering techniques; this enables the researcher to understand the vast possibilities of the kinds of anomaly that a machine can encounter.

Baseline Anomalous Sound Detection System

3.1 Introduction

In this chapter, the baseline anomalous sound detection (ASD) system [7, 3] developed during the research is presented. The main purpose of the baseline system is first, to create a reproducible performance benchmark on the MIMII pump dataset to compare it with the proposed system discussed in the next chapter and second, performing empirical analysis, we showcase the limitations faced when using conventional reconstruction-based approaches. The baseline adopted in the DCASE community is an autoencoder for unsupervised ASD [chalapathy2019deep, 3], which uses normal machine audio to train the model, and the anomalous ones are detected based on how much they deviate from the normal audios.

The approach taken here is a slightly improved version of the conventional practice rather than a state of the art. Here, we implement a convolutional autoencoder [15, 16] based on log-mel spectrogram windows, the approach here is inspired from the original MIMII

benchmark system [2], the baseline acts as a meaningful reference for evaluating the proposed system in the next chapter. The design choices, experimental protocol, and evaluation metrics [17] used in this chapter remain consistent across both systems to compare the performance differences so that we can attribute it to architectural factors rather than evaluation inconsistencies.

3.2 Dataset: MIMII Pump

For all experiments in this chapter we use the MIMII (Malfunctioning Industrial Machine Investigation and Inspection) dataset[2], specifically the pump machine category recorded at a signal-to-noise ratio of 0 dB. The MIMII dataset collected by Hitachi Ltd. [2], is made publicly available for research purposes. In this dataset we have audio files where each of them is a ten-second recording captured using 8 sets of microphones positioned in a circular manner around the target machine in an industrial facility. The dataset contains recordings on four machine IDs (id_00, id_02, id_04, and id_06), each representing a different pump model.

For each machine ID, we have two classes of recordings namely normal, which represents healthy machine operations and the other is abnormal, which represents faulty operations of the machine. The fault types present in the pump category include clogging, leakage, contamination, etc. The abnormal class labels are not introduced during training; the model only learns normal recordings, so that it

can identify anomalies which are deviating from the normal ones at test time. This unsupervised, one-class learning setting is used as in realistic industrial deployment conditions fault examples are either unavailable or are very rare.

The following table is a summarized data composition for machine `id_00`. The same experimental protocol was applied identically to `id_02`, `id_04`, and `id_06`.

Table 3.1: Dataset split for MIMII pump `id_00` (0 dB SNR). The model is trained exclusively on normal files; abnormal files appear only at evaluation time.

Split	Condition	Count	Purpose
Train	Normal only	704	AE training (no labels used)
Validation	Normal	148	Threshold tuning / early stopping
Validation	Abnormal	74	—
Test	Normal	154	Final evaluation (AUC / pAUC)
Test	Abnormal	69	—

3.3 Feature Extraction

3.3.1 Multi-Channel Averaging and Resampling

The MIMII recordings are captured with an eight-channel microphone array; therefore, the first pre-processing step disrupts the spatial dimension by computing the element-wise mean across all eight channels [18]. As a result, we get a mono waveform that retains the ambient acoustic character of the recording while avoiding the added complexity of multichannel fusion architectures. Although previous

work has explored channel selection, the averaging approach is the same as the baseline methodology of the original MIMII paper, so that we can directly compare the results.

We resample all the waveforms to a uniform sampling rate of 16,000 Hz using polyphase resampler [19] where the native sample rate varies. Each waveform is then zero-padded or truncated to a fixed duration of 10 seconds, which gives exactly 160,000 samples per recording.

3.3.2 Log-Mel Spectrogram Computation

The main acoustic representation used by the baseline system is the log-mel spectrogram, which maps the raw waveform into a time-frequency domain that approximates the perceptual sensitivity of the human auditory system [20]. The spectrogram is calculated using a Short-Time Fourier Transform (STFT) [18] followed by mel-filterbank projection and logarithmic compression. The hyperparameters involved in this transformation are summarized in Table 3.2.

Table 3.2: Log-mel spectrogram extraction hyperparameters.

Hyperparameter	Value
Sampling rate (SR)	16,000 Hz
Audio duration	10 seconds
FFT window size (<code>n_fft</code>)	1,024 samples
Hop length	512 samples
Number of mel bins	128
Spectrogram power	1.0 (power spectrum)
Amplitude scaling	Decibel (log) scale, ref = max
Output shape (per file)	128×313 (mel bins $\times$ time frames)

As a result, we get a representation of dimensions 128×313 , which represents the 128 mel-frequency bins and 313 time frames for a ten-second clip at the specified hop length. When converting the decibel we take the maximum power value in each spectrogram, this maps the dynamic range of the signal to a scale that is more manageable to neural network processing so that to reduce the dominance of high-energy impulsive events.

3.3.3 Temporal Windowing

Instead of treating the full 128×313 spectrogram as a single input instance, the proposed baseline system segregates each spectrogram into a sequence of overlapping windows before they are fed to the autoencoder. Here, each window captures a 64-frame temporal slice of the mel spectrogram, which is approximately two seconds of audio at the given hop length. The subsequent windows are offset by 32 frames (50% overlap), yielding approximately eight windows per

ten-second recording. The idea of this windowed representation is to introduce a form of local temporal invariance, such that it substantially augments the effective number of training samples, which greatly benefits us in increasing the data, as the MIMII training set is limiting.

Formally, for a log-mel spectrogram $S \in \mathbb{R}^{M \times T}$ where $M = 128$ and $T = 313$, the windowing operation produces a set of frames $\{S[:, t : t + W]\}$ for $t = 0, W/2, W, \dots$ where $W = 64$. Each window is expanded with a channel dimension to give a tensor of shape $(1 \times 128 \times 64)$, compatible with the two-dimensional convolutional layers of the autoencoder. After windowing, we augment the complete training set (704 normal files) to produce a set containing approximately 7,928 windows.

3.3.4 Normalisation

Before we perform model training, we first normalize the windowed training set by subtracting the per-mel-bin mean and dividing it by the per-mel-bin standard deviation, both computed exclusively over the training windows. Specifically, the mean μ and standard deviation σ are computed by aggregating over both the sample dimension and the time-frame dimension, which produces a normalization vector of length 128 (one scalar per mel bin). This normalization is applied to training, validation, and test windows using statistics derived only from the training set, ensuring that no information leakage

occurs from evaluation splits. The operation also ensures that each mel frequency channel has approximately zero mean and unit variance across the training set, which as a result helps to stabilize the gradient flow during autoencoder training.

3.4 Baseline Model Architecture

We propose a baseline model which is a fully convolutional autoencoder (CAE) that maps a spectral window to a compact latent representation and reconstructs it, learning to capture the structural regularity of only normal machine audio. We use convolutional layers because of the two-dimensional nature of the input: the mel spectrogram has meaningful structure along both the frequency axis (harmonic content, spectral shape) and the time axis (temporal patterns, periodicity), and convolutional operations can use both without the parameter overhead of fully connected architectures.

3.4.1 Encoder

The encoder we use in the system has four successive convolutional blocks, where each of the blocks has a 3×3 convolution with stride 2, a Batch Normalization [21] layer and a ReLU activation. With the stride of 2 in each block we reduce the spatial resolution by half, so together the four blocks reduce the 128×64 input to an 8×4 spatial map. The number of convolutional channels increases progressively

as we go through the network: $1 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128$, this enables the model to build abstract representations at increasingly coarser temporal and spectral resolutions. The final output of the encoder as a result is a tensor of shape $128 \times 8 \times 4$, made up of the latent part from which the audio is reconstructed. Batch Normalization is applied after each convolution (before the non-linearity) to help accelerate training convergence and as a result reduce sensitivity to initialization.

3.4.2 Decoder

In the decoder, we just mirror the encoder structure by applying four transposed convolution blocks to increase up-sampling of the latent representation back to the original $1 \times 128 \times 64$ shape. Each transposed convolution uses a 3×3 kernel with stride 2 and output padding 1, and is followed by a ReLU activation (without Batch Normalization, as used in standard autoencoder practice in the decoder). The channel sequence reverses what was done in the encoder: $128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 1$. A spatial crop is applied to the final decoder output as it removes any off-by-one padding artifacts which got introduced due to the transposed convolution, which ensures that the shape of the reconstructed window matches exactly with the shape of the input window. No activation function is applied to the final layer, this allows the reconstruction to span the full range of the (normalized) input values.

3.4.3 Architecture Summary

Table 3.3: Layer-wise architecture of the convolutional autoencoder baseline.

Stage	Operation	Output Shape	BN	Actv. Func
Input	—	$1 \times 128 \times 64$	—	—
Enc-1	Conv2d($1 \rightarrow 16$, 3×3 , s=2)	$16 \times 64 \times 32$	Yes	ReLU
Enc-2	Conv2d($16 \rightarrow 32$, 3×3 , s=2)	$32 \times 32 \times 16$	Yes	ReLU
Enc-3	Conv2d($32 \rightarrow 64$, 3×3 , s=2)	$64 \times 16 \times 8$	Yes	ReLU
Enc-4 (Latent)	Conv2d($64 \rightarrow 128$, 3×3 , s=2)	$128 \times 8 \times 4$	Yes	ReLU
Dec-1	ConvTranspose2d($128 \rightarrow 64$, 3×3 , s=2)	$64 \times 16 \times 8$	No	ReLU
Dec-2	ConvTranspose2d($64 \rightarrow 32$, 3×3 , s=2)	$32 \times 32 \times 16$	No	ReLU
Dec-3	ConvTranspose2d($32 \rightarrow 16$, 3×3 , s=2)	$16 \times 64 \times 32$	No	ReLU
Dec-4 (Output)	ConvTranspose2d($16 \rightarrow 1$, 3×3 , s=2) + Crop	$1 \times 128 \times 64$	No	None

3.5 Training Procedure

Using the Adam optimizer [22], we train the autoencoder with a fixed learning rate and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). We use the Mean Squared Error (MSE) as the loss function between the input window and its reconstructed output, averaged over all spatial elements. Then we train the model with a batch size of 64. Since the training set contains only normal recordings, the model never learns abnormal samples during training; therefore, it is optimized solely to minimize the normal reconstruction error.

Model selection is performed using early stopping on the basis of validation loss. At each epoch, the total validation MSE (com-

puted over the mixed normal and abnormal validation set, but with reconstruction loss computed to labels) is observed and the model weights having the lowest validation loss are saved. As a result, we prevent the model from overfitting the training set and ensure that the retained model can generalize unseen acoustic conditions. The validation set contains both normal and abnormal files; note that the validation loss does not speak for AUC, since the autoencoder is unsupervised and has no access to labels, the validation set here serves only as a held-out sample from the overall data to perform regularization.

we observed a stable training convergence across all four machine IDs. For `id_00`, the training loss decreased from approximately 0.746 at epoch 10 to approximately 0.202, the validation loss reaching approximately 0.157, from which we can say that the model did not significantly overfit to the training windows. Similar convergence patterns were observed across the remaining machine IDs.

3.6 Anomaly Scoring

For inference, we calculate an anomaly score for each ten-second audio file by first segregating it into multiple overlapping windows using the same parameters as during training (window size 64 frames, hop 32 frames), we then normalize each window using the training-set statistics (μ, σ) , and pass each window through the trained autoencoder. The per-window anomaly score is the Mean Squared Error

between the input window and its reconstruction:

$$\text{score}_{\text{window}} = \frac{1}{M \times W} \sum_{i,t} (x_{i,t} - \hat{x}_{i,t})^2 \quad (3.1)$$

where W is the window size, M is hop frames, $x_{i,t}$ is the normalized input and $\hat{x}_{i,t}$ is the reconstructed output in the mel bin i and the time frame t . The anomaly score is then obtained by averaging the per-window scores across all windows extracted from the audio file:

$$\text{score}_{\text{file}} = \frac{1}{N_w} \sum_k \text{score}_{\text{window},k} \quad (3.2)$$

where N_w denotes the number of windows extracted from the audio file. A higher score indicates that the recording shows a strong deviation from the normal acoustic patterns which the model learned during training, and is therefore more likely to be anomalous. This score is used as input for the ROC-based evaluation metrics described in the following section, without performing any further post-processing or thresholding calibration.

3.7 Evaluation Protocol and Metrics

For evaluation, we follow the protocol established by the DCASE 2020 Challenge Task 2 [3], since it is the primary benchmark related with the MIMII dataset [2].

The Area Under the ROC Curve (AUC-ROC) [23] measures the probability that a randomly selected anomalous audio data gives

a higher anomaly score as compared to a randomly selected normal audio data. Usually the AUC-ROC ranges from 0.5 (chance) to 1.0 (perfect separation) [chalapathy2019deep] which is independent of threshold, this makes it the most appropriate metric for the unsupervised detection setting where we do not have validation-set threshold prior to final evaluation. The metrics are calculated using the anomaly scores described in Section 3.6.

3.8 Experimental Results

3.8.1 Performance Comparison Across Machine IDs

Table 3.4 presents the AUC-ROC scores of the proposed baseline convolutional autoencoder and the MIMII Paper system reported on all four machine IDs of pump at 0 dB SNR. The result of reconstruction-based anomaly detection on log-mel spectrograms shows moderate but inconsistent performance across machine units, which is consistent with prior reports in the DCASE community. The original MIMII paper by Purohit et al. (2019) [2] introduced a baseline system using a fully connected autoencoder operating on log-mel spectrograms with 640-dimensional input vectors (concatenating five consecutive frames of 128 mel bins each). In this table, we compare the results of the convolutional autoencoder obtained in this chapter against the published baseline numbers for the pump category at 0 dB SNR.

Table 3.4: Baseline convolutional autoencoder vs. MIMII paper (pump, 0 dB SNR).

Machine ID	Baseline CAE (AUC-ROC)	MIMII (AUC-ROC)
id_00	0.74	0.65
id_02	0.44	0.46
id_04	0.98	0.95
id_06	0.74	0.76
Average	0.73	0.70

From this table we can see that even within the same machine category and noise condition, both the system’s ability to discriminate normal from anomalous sounds varies significantly across machine IDs. This variability is caused because each machine ID represents a physically different unit which has its own characteristic operational audio; the autoencoders that are trained on one unit are not transferable across other units; also the fault acoustic feature differ across the units as well. This shows that the above systems are sensitive to the type of unit used, which is one of the limitations addressed by the proposed system in Chapter 4.

The use of the convolution architecture enables the system to use the two-dimensional spatial structure of the mel spectrogram through parameter-efficient shared filters, rather than treating each spectrogram element as an independent feature.

3.9 Analysis and Discussion

- **Reconstruction Error is an Unreliable Indicator:** The autoencoder often reconstructs anomalous sounds reasonably well because many anomalies share similar acoustic patterns with normal data. This reduces the contrast in reconstruction error, making anomalies harder to detect.
- **Limited Expressiveness of Handcrafted Features:** Log-mel spectrograms do not capture deeper contextual or semantic audio patterns, which restricts the model's ability to distinguish subtle faults.
- **Lack of Structured Latent Space:** The latent representations are not explicitly constrained, resulting in poorly defined boundaries between normal and anomalous data.
- **Sensitivity to Machine Variability:** Performance varies significantly across different machine units, indicating limited generalization capability of the model.
- **Weak Fault Interpretability:** The system only detects anomalies but does not provide insight into the type or structure of the fault, limiting its usefulness for practical diagnosis.

From the results we obtained in this chapter, we can see that there are several structural limitations of this improved baseline ASD system, which makes it crucial to understand before we introduce the proposed system in the next chapter. These limitations are not the

result of evaluation inconsistencies; they are present in the underlying approach.

Proposed System: AST with Complete Deep SVDD

4.1 Introduction and Motivation

The baseline system we introduced previously established that a convolutional autoencoder operating on log-mel spectrograms achieves moderate anomalous sound detection performance on the MIMII pump dataset, but exposes three structural limitations that constrain its ceiling: the weakness of reconstruction error as an anomaly proxy, the shallowness of hand-engineered mel spectrogram features, and the absence of geometric structure in the latent space. This chapter introduces the proposed system designed to address each of these limitations through two main architectural innovations.

The first innovation replaces the log-mel spectrogram feature extractor with the **Audio Spectrogram Transformer (AST)** [6], a large-scale model pretrained on AudioSet [24] that provides rich, semantically meaningful acoustic embeddings through a vision transformer [25, 26] operating directly on the spectrogram. The second innovation replaces the reconstruction-based anomaly score with

a **Complete Deep Support Vector Data Description (CD-SVDD)** [27] objective, which explicitly trains a neural network to map normal samples to a compact hypersphere in the embedding space, yielding a principled geometrically motivated anomaly score [26]. The two components are integrated into a unified three-stage pipeline: AST fine-tuning via masked spectrogram prediction, embedding extraction, and alternating optimization of the CD-SVDD network and its hypersphere parameters. The result is a system that addresses the fundamental limitations of reconstruction-based ASD whilst remaining fully unsupervised, no anomaly labels are required at any stage of training.

4.2 System Overview

The proposed pipeline consists of three sequential stages, each building on the output of the previous:

Stage 1: AST Fine-tuning via Masked Spectrogram Prediction. The pretrained AST is adapted to the acoustic domain of MIMII pump recordings using a self-supervised masked patch reconstruction objective. Only the last four transformer encoder blocks are unfrozen; the remaining layers retain their AudioSet-pre-trained weights. This stage updates the model representations $\mathbf{l}$ s to better reflect the spectral characteristics of the audio of industrial machinery without requiring labeled data.

Stage 2: Embedding Extraction. The fine-tuned AST is used as a fixed feature extractor. Each ten second audio file is processed to produce a 1536-dimensional embedding. The embeddings are standardized using statistics computed from the normal training set and normalized to L2 before being passed to the SVDD stage.

Stage 3: Complete Deep SVDD Training. A three-layer feedforward network is trained to map normalized AST embeddings to a 128-dimensional hypersphere. Training follows the CD-SVDD formulation with alternating optimization: the network weights are updated by gradient descent on the primal objective while the hypersphere center and radius are updated by solving the dual Quadratic Programming (QP) problem. The final anomaly score for a test recording is its squared Euclidean distance from the learned hypersphere center in the 128-dimensional latent space [28].

4.3 Audio Spectrogram Transformer (AST)

4.3.1 Architecture and Pretraining

The Audio Spectrogram Transformer (AST) [6] is a fully attention-based model for audio classification that treats the log-mel spectrogram as a sequence of two-dimensional patch embeddings, in analogy with the Vision Transformer (ViT) [26]. Given a log-mel spectrogram of shape $(T \times F)$, the AST divides it into overlapping patches of fixed size using a stride-based convolutional patch embedding layer,

projects each patch to a 768-dimensional token vector, and prepends a learnable classification token [CLS]. The sequence of patch tokens is then processed by a stack of 12 multi-head self-attention transformer encoder blocks. The model is pretrained on AudioSet [24], a large-scale audio dataset containing over 2 million ten-second clips spanning 527 audio event classes, making its representations acoustically informed in a way that mel-spectrogram features computed from scratch are not.

This work uses a pretrained model, which is the publicly available AST finetuned on the full AudioSet balanced and unbalanced sets with patch overlap of 10×10 pixels, achieving a mean average precision of 0.4593 on AudioSet. The AST's feature extractor is used to transform raw waveforms into the model's expected input format (a 128-bin log-mel spectrogram of fixed length), ensuring consistent preprocessing across all audio files.

4.3.2 Embedding Dimensionality

Rather than using only the [CLS] token output (768 dimensions), the embedding for each audio file is constructed by concatenating the [CLS] token representation with the mean-pooled representation of all patch tokens from the final transformer layer, yielding a **1536-dimensional** embedding vector. This concatenation strategy preserves both the global audio summary encoded in the [CLS] token and the averaged local spectral structure encoded across all patch

positions, providing a richer input to the SVDD network than either representation alone. Empirically, this approach has been shown to produce more stable SVDD training compared to using the 768-dimensional CLS-only embedding, as the increased dimensionality provides a richer basis from which the network can learn a compact normal manifold.

4.3.3 Domain Adaptation via Masked Spectrogram Prediction

Although the AST is pretrained on AudioSet, the acoustic characteristics of industrial pump sounds differ substantially from the diverse audio events in AudioSet's training distribution. Directly applying the pretrained AST without adaptation risks producing embeddings that capture AudioSet-relevant semantic features but are insensitive to the subtle spectral differences that distinguish normal from anomalous pump operation. To address this, the AST is fine-tuned on the MIMII pump training set using a **masked spectrogram prediction** (MSP) objective, a self-supervised reconstruction task analogous to masked language modelling in NLP.

During fine-tuning, a fraction $r = 0.4$ of the non-CLS patch tokens are randomly masked at each forward pass. The fine-tuner appends a two-layer reconstruction head (Linear $\rightarrow$ GELU $\rightarrow$ Linear, mapping $768 \rightarrow 768$) to the AST output and is trained to minimise the Mean Squared Error between the predicted and actual representations of

the masked patches. All AST layers are frozen during fine-tuning **except the last four transformer encoder blocks**, which are unfrozen and updated with gradients from the MSP loss. This partial unfreezing strategy balances adaptation to the target domain against the risk of catastrophic forgetting of the broad acoustic knowledge encoded in the earlier layers.

Table 4.1: AST fine-tuning configuration for the masked spectrogram prediction stage.

Configuration	Value
Base model	MIT/ast-finetuned-audioset-10-10-0.4593
Frozen layers	Transformer blocks 1–8 (of 12)
Unfrozen layers	Transformer blocks 9–12 + patch embed
Mask ratio	0.4 (40% of patches masked per pass)
Reconstruction target	Masked patch representations (MSE)
Optimiser	AdamW
LR schedule	Cosine Annealing ($T_{\max} = n_{\text{epochs}}$)
Training epochs	10
Batch size	16 files
Avg val loss	0.00105

Fine-tuning converges rapidly given the large pretrained capacity of the AST. For pump id_00, the training reconstruction loss decreases from an initial value to 5.89×10^{-4} after 10 epochs, with a validation loss of 1.17×10^{-3} , indicating stable adaptation without overfitting. The fine-tuned weights are saved per machine ID and used exclusively for embedding extraction in the subsequent stage.

4.4 Embedding Extraction and Preprocessing

Following fine-tuning, the AST is used as a fixed feature extractor with frozen weights. Each audio file is loaded at 16,000 Hz and passed through the pretrained feature extractor to produce the model's standard 128-bin log-mel spectrogram input. The fine-tuned AST then processes this input and returns the **1536-dimensional** embedding (CLS token concatenated with mean-pooled patch tokens), one vector per audio file. Embeddings are extracted separately for the training set (normal files only), the validation set (mixed), and the test set (mixed).

The extracted embeddings undergo two sequential preprocessing steps before being passed to the SVDD network. First, we standardize each of the 1536 dimensions to zero mean and unit variance. Second, L2 normalization is applied to project each embedding vector onto the unit hyper-sphere. This normalization step is important for CD-SVDD training: by constraining all embeddings to the surface of a fixed-radius sphere in input space, the optimization landscape for the hyper-sphere center and radius in the 128-dimensional latent space becomes more well-conditioned, and the QP dual solutions are more numerically stable.

4.5 Complete Deep Support Vector Data Description

4.5.1 Motivation and Theoretical Background

Deep Support Vector Data Description [27] is a one-class classification method that trains a neural network $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ to map normal training samples into a compact hypersphere centred at $c \in \mathbb{R}^k$ with radius $R \in \mathbb{R}$ in the latent space. The standard Deep SVDD objective minimises:

$$\min_{W,R} R^2 + \frac{1}{\nu N} \sum_i \max(0, \|\phi(x_i; W) - c\|^2 - R^2)$$

where $\nu \in (0, 1]$ is a hyperparameter controlling the trade-off between the hypersphere volume and the penalty for samples falling outside it, N is the number of training samples, and W denotes the network parameters. At inference time, the anomaly score for a test sample x is its squared distance from the centre: $s(x) = \|\phi(x; W) - c\|^2$. Higher scores indicate greater deviation from the learned normal distribution.

A critical vulnerability of naive Deep SVDD is *hypersphere collapse*: if the network is unconstrained, it can trivially minimise the objective by mapping all inputs to a single constant (e.g., the zero vector), yielding a degenerate solution. The standard safeguards are the removal of all bias terms and the use of Batch Normalisation without learnable affine parameters (**affine=False**), both of which

are implemented in the proposed system's SVDD network.

The **Complete Deep SVDD** formulation (used in this work) extends standard Deep SVDD by jointly optimising the network weights and the hypersphere parameters (centre c and radius R) through an alternating optimisation strategy grounded in the dual formulation of the SVDD problem. Rather than fixing the centre as the mean of the initial network outputs and updating only the weights, CD-SVDD iteratively re-estimates both c and R via the dual QP solve, providing a theoretically tighter approximation to the minimal enclosing hypersphere of the normal training embeddings.

4.5.2 Network Architecture

The SVDD network ϕ is a three-layer feedforward network that maps the 1536-dimensional normalised AST embedding to a 128-dimensional latent representation. The architecture follows the encoder of the pretraining autoencoder (described in Section 4.5.3). Each hidden layer applies a **bias-free linear transformation**, followed by **Batch Normalisation with affine=False**, and a ReLU activation. No activation is applied after the final projection layer, which outputs the 128-dimensional latent code. The channel sequence is:

Table 4.2: Architecture of the Deep SVDD network ϕ . All linear layers omit bias terms to prevent hypersphere collapse.

Layer	Operation	Output Dim	Notes
Input	—	1536	Normalised AST embedding
FC-1	Linear(1536→256, bias=False)	256	BN (affine=False) + ReLU
FC-2	Linear(256→128, bias=False)	128	BN (affine=False) + ReLU
Output	Linear(128→128, bias=False)	128	No activation (latent code)

4.5.3 Three-Stage Training Protocol

Training the CD-SVDD network proceeds in three sequential stages, designed to ensure stable initialisation and avoid the degenerate solutions that can arise from random weight initialisation in standard Deep SVDD.

Stage 3a: Autoencoder Pretraining.

A symmetric autoencoder sharing the SVDD network's encoder architecture is first trained on the normal training embeddings using MSE reconstruction loss for 50 epochs with the Adam optimiser. The decoder mirrors the encoder: $128 \rightarrow 128 \rightarrow 256 \rightarrow 1536$, using the same bias-free linear layers and affine-free Batch Normalisation. This pretraining phase serves as a form of weight initialisation: the encoder is forced to learn a compact representation of the normal AST embedding distribution before the SVDD objective is introduced. The autoencoder training converges from a loss of

approximately 9.75×10^{-3} at epoch 10 to 7.43×10^{-4} at epoch 50 for id_00, indicating effective reconstruction of the 1536-dimensional embedding space.

Stage 3b: Encoder Weight Transfer.

Following autoencoder pretraining, the encoder weights are transferred layer-by-layer to the SVDD network ϕ by matching parameter names and shapes between the autoencoder's encoder state dictionary and the SVDD network's state dictionary. This transfer initialises the SVDD network in a region of parameter space where the mapping from AST embeddings to the 128-dimensional latent space already captures meaningful structure, significantly stabilising the subsequent CD-SVDD training.

Stage 3c: CD-SVDD Alternating Optimisation.

The CD-SVDD training phase runs for 100 epochs with the SGD-based update rule (learning rate 0.1) on the primal SVDD objective. The hypersphere centre c is not set during the first 10 epochs (warm-up phase), allowing the network to first develop a stable latent representation before the compact hypersphere constraint is imposed. At epoch 10, the centre c is initialised by solving the dual QP over the full training set, obtaining the support vectors and their dual coefficients. Thereafter, at each epoch, the network weights are updated by gradient descent on the primal loss while the centre and radius are re-estimated from the current latent representations by solving:

$$\min_{\alpha} \alpha^{\top} Q \alpha - \alpha^{\top} \text{diag}(Q) \text{ s.t. } \sum_i \alpha_i = 1, 0 \leq \alpha_i \leq \frac{1}{\nu \cdot l}$$

where $Q_{ij} = \phi(x_i)^{\top} \phi(x_j)$ is the Gram matrix of latent representations and l is the number of training samples in the current batch. The dual solution α^* is obtained using Sequential Least Squares Programming (SLSQP) with box and equality constraints. The optimal centre is recovered from the dual solution as the weighted sum of support vectors, and the radius is set to the maximum distance from the centre over non-outlier training samples. This alternating scheme provides tighter hypersphere estimates than the fixed-center approximation used in standard Deep SVDD, at the cost of a dual QP solve per epoch.

Table 4.3: Complete Deep SVDD hyperparameters.

Hyperparameter	Value
Input dimension	1536 (AST embedding)
Hidden dimensions	[256, 128]
Latent dimension (k)	128
ν (nu)	0.3
AE pretraining epochs	50
SVDD training epochs	100
Warm-up epochs	10 (centre uninitialised)
Optimiser (SVDD)	SGD (lr = 0.1)
Dual QP solver	SLSQP (scipy.optimize)
Batch size	64 embeddings
Anomaly score	$\ \phi(x; W) - c\ ^2$ (squared distance)

4.6 Evaluation Protocol

The evaluation protocol for the proposed system is identical to that of the baseline in Chapter 3, ensuring direct comparability. Both

AUC-ROC [23] and pAUC ($p = 0.1$) are reported for each machine ID using the same train/validation/test split and the same data composition. The anomaly score assigned to each test file is the squared Euclidean distance of its 128-dimensional SVDD latent representation from the learned hypersphere centre c . No threshold is required for ROC-based evaluation; the threshold-dependent accuracy metric reported in Table 4.5 uses Youden's J statistic [17] (optimal point on the validation ROC curve) to calibrate the decision boundary.

Model selection during SVDD training uses the validation AUC-ROC, computed at every epoch over the mixed validation set, as the criterion for saving the best checkpoint. This is notably different from the baseline, where validation MSE loss was used for early stopping; AUC-based model selection is more directly aligned with the evaluation objective and is made possible by the availability of mixed validation labels in both systems. The best checkpoint weights are loaded for final test evaluation.

4.7 Experimental Results

4.7.1 Proposed System Performance Across Machine IDs

Table 4.4 presents the AUC-ROC and pAUC scores of the proposed AST + CD-SVDD system across all four pump machine IDs at 0 dB SNR. The best validation AUC observed during SVDD training for id_00 was **0.8224**, indicating that the system learned to distinguish

normal from anomalous embeddings on the validation set. Final test evaluation numbers for all IDs should be filled from your evaluation runs.

Table 4.4: Performance metrics across pump machine IDs.

Machine ID	Current AUC-ROC	MIMII AUC-ROC
pump/id_00	0.82	0.65
pump/id_02	0.97	0.46
pump/id_04	0.95	0.95
pump/id_06	0.91	0.76
Average	0.91	0.71

4.7.2 Comparison with the Baseline System

Table 4.5 presents a head-to-head comparison of the proposed system against the baseline convolutional autoencoder from Chapter 3 and the original MIMII paper baseline, for the pump category at 0 dB SNR. The comparison is made under identical experimental conditions: same audio files, same train/val/test split, same evaluation metrics.

Table 4.5: System comparison on MIMII pump (0 dB SNR). Gain reported relative to the MIMII baseline.

System	Mean AUC-ROC	↑ Gain (%)
MIMII Paper (FC-AE, 0 dB)	0.71	—
Baseline CAE (Chapter 3)	0.73	2.82
Proposed: AST + CD-SVDD (Ours)	0.91	28.17

Here we see massive improvement in the AUC-ROC scores for the

proposed system and slight improvement in the proposed Baseline system as compared to the system proposed in the MIMII paper. This tells us that the AST+CD-SVDD system captures the spectral features of the audio far efficient with high precision, compactness and stability, as a result shows a proper segregation between normal and anomalous audio data.

4.8 Discussion

- **Limited Noise Condition Evaluation:** The system is evaluated under a single noise setting, which does not fully capture real-world variability in acoustic environments.
- **Training Stability of SVDD:** The alternating optimisation process can introduce fluctuations during training, affecting consistency across epochs.
- **Computational Overhead:** The use of transformer-based feature extraction increases computational cost, especially for large-scale or real-time deployment.
- **Dataset and Domain Dependency:** The model is trained and evaluated on a specific machine category, which may limit its generalisation to other industrial settings without adaptation.
- **Lack of Real-Time Deployment Validation:** The current work focuses on offline evaluation, and system performance in a

real-time monitoring pipeline remains to be validated.

Post-Hoc Fault Subtype Discovery via Clustering

5.1 Introduction and Motivation

In the previous chapters, we solved the detection problem, where given an audio recording of an industrial pump, we try to determine whether its operating condition is normal or anomalous. In the context of predictive maintenance, detection is only the first layer. Another aspect of it is the kind of anomaly that has been detected. Based on how we distinguish between, for example, a clogging fault and a voltage irregularity, there are many direct consequences for maintenance scheduling, spare-parts provisioning, and root-cause analysis. If anomalous recordings from the same machine unit cluster into acoustically coherent groups [12] in the latent space of the proposed detection system, that structure can be leveraged to get an idea of the fault subtypes, entirely without any fault-type labels.

In this chapter, we perform a post-hoc analysis of the discovery of fault subtypes on the anomalous audio samples of the MIMII pump

dataset (0 dB SNR). The analysis is post-hoc in nature, as it operates on the outputs of the trained models from the previous chapters rather than modifying or creating those systems from scratch for the analysis; it is exploratory in nature, rather than being a supervised classification. Here we apply two complementary clustering approaches: HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) as the primary method, and Deep Embedded Clustering (DEC) [14] with generalizability-based cluster count selection as the secondary method. We perform analysis on two independent embedding spaces to check whether we can get actual acoustic structure, one on the fine-tuned AST embeddings from Chapter 4, and then on CAE bottleneck embeddings from the Chapter 3 baseline, we also make note of agreement between methods and embeddings, quantifying it using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

5.2 Data and Embedding Spaces

5.2.1 Anomalous Sample Set

We perform analysis on the anomalous test recordings of all four IDs of the pump machine. The anomalous sample counts are 143 files for id_00, 111 for id_02, 100 for id_04, and 102 for id_06 of the MIMII dataset at 0 dB SNR. Normal recordings are not included; the clustering is applied only to anomalous audio data, since we are

investigating the internal structure and spectral feature *within* the anomaly class rather than discriminating anomalies from normals.

5.2.2 AST Embedding Space

The fine-tuned AST from Chapter 4 is used as the primary embedding space. Each anomalous recording is passed through the fine-tuned AST feature extractor and model resulting in a **1536-dimensional** embedding per file (CLS token concatenated with mean-pooled patch tokens), exactly as in the Chapter 4 training pipeline. The AST weights are the fine-tuned checkpoints per machine saved during Chapter 4 training; no further adaptation is performed. The resulting embedding matrices have shapes (143×1536) , (111×1536) , (100×1536) , and (102×1536) for id_00 through id_06, respectively.

5.2.3 CAE Embedding Space

The secondary embedding space uses the convolutional autoencoder encoder from Chapter 3 as a feature extractor applied to the same anomalous audio data. For each of the files, a log-mel spectrogram is created having the same parameters as used in Chapter 3 ($\text{SR} = 16,000 \text{ Hz}$, $\text{n_mels} = 128$, $\text{n_fft} = 1024$, $\text{hop} = 512$), which are then passed through the trained CAE encoder, and the spatial output is flattened to yield a **20,480-dimensional** embedding per file. This high-dimensional representation captures the structure learned when reconstructing the normal audio data and serves as the independent

comparison basis for the AST embeddings. Using the CAE encoder features instead of the reconstruction error scalars, we can do spatial clustering over the learned representation rather than over a single anomaly score.

5.2.4 Dimensionality Reduction Pipeline

Before clustering is performed, identical preprocessing is performed for both types of embedding. First, it is normalized. Then a dimensionality reduction technique, Principal Component Analysis (PCA), is applied to retain 95% of the explained variance while preserving the principal structure. For the AST embeddings, this produces a compact representation suitable for density estimation; for the CAE embeddings, which start at 20,480 dimensions, PCA reduction is essential to make the density-based methods computationally efficient. For plotting cluster assignments, UMAP [mcinnes2018umap, 29] (`n_neighbors = 15`, `min_dist = 0.1`) is additionally applied to PCA-reduced representations and is not used as input to the clustering algorithms.

5.3 Primary Method: HDBSCAN Clustering

5.3.1 Algorithm and Parameter Choice

HDBSCAN [13, 30] is a hierarchical density-based clustering algorithm that creates a hierarchy of density-connected clusters and creates a flat partition by optimizing cluster stability. Its key advantages for this application are: (1) it does not require the number of clusters k to be known in advance, it is discovered from the data density; (2) rather than forcing every point into a cluster, it explicitly handles noise by assigning low-density points the label -1 ; and (3) it assigns a stability score to each discovered cluster, providing an internal measure of cluster robustness. These properties are well-suited to the unsupervised character of fault subtype discovery, where the number and prevalence of fault types is unknown.

5.3.2 HDBSCAN Results for pump id_00

Table 5.1: HDBSCAN results on AST and CAE embeddings for pump id_00 (143 anomalous samples). Cluster stabilities are HDBSCAN’s internal persistence scores; higher is more stable.

Metric	AST	CAE	AST Cluster Stabilities	CAE Cluster Stabilities	N
Clusters found	3	2	0.162 / 0.176 / 0.108	0.251 / 0.013	143
Noise fraction	0.035 (3.5%)	0.916 (91.6%)	—	—	
Silhouette Score	0.4058	0.3973	—	—	
Davies-Bouldin Index	1.0266	0.9266	—	—	

There is a massive difference between the two embedding spaces.

In the AST embeddings, HDBSCAN discovers 3 clusters with a noise fraction of only 3.5%: nearly all anomalous samples are assigned to a cluster. The cluster stabilities (0.162, 0.176, 0.108) are modest but comparable, indicating that the three clusters are similarly well-supported by the local density structure. The Silhouette Score [31] of 0.4058 indicates moderate but non-trivial separation between the discovered groups.

In the CAE embeddings, the result is quite different: HDBSCAN assigns 91.6% of samples as noise, retaining only a small minority in two clusters. The second CAE cluster has a stability of just 0.013, which is essentially indistinguishable from noise. This reason these findings are subpar is not due to HDBSCAN; it is due to the result of the CAE embedding geometry. Since the CAE encoder is trained to minimize reconstruction error on normal training windows, it produces representations which cannot discriminate over different types of anomalous sounds, and since anomalous samples are scattered diffusely in the CAE latent space, without the density concentrations, HDBSCAN fails to identify clusters. This reflects the limitation identified in Chapter 3: the CAE’s latent space encodes reconstruction-relevant features of normal audio, not semantically meaningful acoustic structure for anomalous conditions.

5.3.3 Spectral Characterisation of AST Clusters (id_00)

To assess whether the three clusters discovered by HDBSCAN in the AST embedding space correspond to acoustically distinct fault subtypes, spectral descriptors are computed for each cluster member and summarized per cluster. Table 5.2 presents the mean and standard deviation of four descriptors, spectral centroid, spectral bandwidth, spectral rolloff and RMS energy, for each of the three AST clusters for id_00.

Table 5.2: Spectral descriptor statistics per HDBSCAN cluster on AST embeddings, pump id_00. Computed over all non-noise anomalous samples assigned to each cluster.

Cluster	Centroid (Hz)	Bandwidth (Hz)	Rolloff (Hz)	RMS	Interpretation
0	1652.85 ± 222.96	1755.52 ± 124.56	3433.22 ± 465.06	0.01 ± 0.00	Higher freq. content, wider bandwidth — possible high-freq fault
1	1426.88 ± 106.95	1532.29 ± 82.48	2788.68 ± 268.29	0.01 ± 0.00	Lower centroid, tighter bandwidth — spectrally compact anomaly
2	1353.75 ± 171.45	1668.31 ± 190.27	2832.52 ± 452.13	0.01 ± 0.00	Lowest centroid, moderate bandwidth — possible low-freq fault

The three clusters show a consistent ordering along the spectral centroid axis: Cluster 0 (centroid ≈ 1653 Hz) $>$ Cluster 1 (≈ 1427 Hz) $>$ Cluster 2 (≈ 1354 Hz). The spectral rolloff also follows the same ordering ($3433 > 2789 > 2833$ Hz respectively for Clusters 0, 1, 2), and the bandwidth differences are also meaningful: Cluster 1 is notably tighter in bandwidth (1532 Hz) compared to Clusters 0 and 2, showcasing a more spectrally concentrated fault signature. These differences are consistent with the moderate Silhouette Score of 0.4058, the clusters are real but not crisply separated in spectral

feature space. Importantly, the RMS energy is essentially identical across clusters (0.01 for all), which indicates that the discovered structure is a result of spectral shape differences rather than amplitude differences [32].

5.4 Secondary Method: Deep Embedded Clustering with Generalizability-Based k Selection

5.4.1 Motivation for DEC

Although HDBSCAN is well-suited to discover clusters from the density structure when we do not know the number of clusters k , it is a density-based geometric method that does not update the embedding space jointly with the clustering objective. Deep Embedded Clustering (DEC; Xie et al., 2016) simultaneously learns a low-dimensional embedding and a soft cluster assignment by repeatedly refining cluster centroids using the Kullback-Leibler divergence between a soft assignment distribution and a sharpened target distribution. The DEC embedding is therefore specifically adapted to separate the data into clusters, whereas the HDBSCAN solution operates on the fixed embedding that is given. The two methods impose structurally different inductive biases, this helps identify which type of clustering provides the robustness of the discovered clusters based on their agreement (or disagreement).

5.4.2 Generalizability-Based k Selection

Since DEC requires that the number of clusters k be specified in advance, we select it based on the data. The **generalizability score** for a given k is defined as the mean nearest-centroid reconstruction error of held-out samples across bootstrap resamples of the training data: for each of 20 bootstrap iterations, K-Means is fitted on 80% of the embeddings, and the mean distance from the held-out 20% to their nearest centroid is recorded. The k that minimizes this mean out-of-sample error is selected as the optimal cluster count, favoring solutions that generalize robustly rather than those that overfit to the specific sample arrangement. This criterion is applied to the range $k \in \{2, 3, 4, 5, 6, 7, 8\}$ for both embedding types.

The generalizability criterion selects $k^* = 4$ for id_00 and $k^* = 5$ for id_02, id_04, and id_06 on the AST embeddings, reflecting genuine inter-machine variation in the number of acoustically distinct anomaly subtypes present in each unit’s test recordings.

5.4.3 DEC Architecture and Training

The DEC model consists of a deep autoencoder with a stacked encoder architecture ($\text{input_dim} \rightarrow 512 \rightarrow 256 \rightarrow 64$) and a symmetric decoder. Batch Normalisation and ReLU activations are applied after each hidden layer. A soft assignment clustering layer appended to the 64-dimensional encoder output computes Student t-distribution

similarities between each embedded point and the k learnable cluster centroids:

$$q_{ij} = \frac{(1 + \|z_i - \mu_j\|^2/\alpha)^{-(\alpha+1)/2}}{\sum_{j'} (1 + \|z_i - \mu_{j'}\|^2/\alpha)^{-(\alpha+1)/2}}$$

where z_i is the embedded point, μ_j is the j -th centroid, and $\alpha = 1$ (degrees of freedom). Training proceeds in two phases: (1) autoencoder pretraining for 100 epochs with MSE reconstruction loss, followed by K-Means initialization of the cluster centroids on the pre-trained encoder’s outputs; (2) DEC clustering refinement for 150 epochs, minimizing the KL divergence between soft assignments q_{ij} and the sharpened target distribution $p_{ij} \propto q_{ij}^2 / \sum_i q_{ij}$.

Table 5.3: DEC architecture and training configuration.

DEC Hyperparameter	Value
Encoder dims	input_dim $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 64
Latent dim	64
k^* (AST, id_00)	4 (generalizability-selected)
k^* (AST, id_02/04/06)	5 (generalizability-selected)
Pretraining epochs	100
Clustering epochs	150
Centroid init	K-Means on pretrained encoder output
Optimiser	Adam (lr = 1×10^{-3})
Batch size	32
Cluster loss	KL($P \parallel Q$) divergence

Table 5.4: Internal clustering validity scores for DEC and HDBSCAN across both embedding spaces, pump id_00.

Metric	DEC / AST	DEC / CAE	HDBSCAN AST	/ HDBSCAN CAE
Clusters (k)	4	4	3	2
Silhouette Score	0.3872	0.0596	0.4058	0.3973
Davies-Bouldin	1.0325	5.1139	1.0266	0.9266

5.4.4 DEC Results for pump id_00

On the AST embeddings, DEC has a Silhouette Score of 0.3872 and a Davies-Bouldin Index of 1.0325, values closely comparable to HDBSCAN’s 0.4058 and 1.0266, respectively. This close correspondence in internal validity between the two fundamentally different methods (density-based vs. distribution-matching) on the same embedding space is notable and suggests that the cluster structure in the AST embedding space is genuinely present.

On the CAE embeddings, DEC’s Silhouette Score comes down to 0.0596 with a Davies-Bouldin Index of 5.1139, a dramatic deterioration relative to both DEC/AST and HDBSCAN/CAE (which had a silhouette of 0.3973 on the small fraction of non-noise points). DEC clustering effectively forces $k = 4$ assignments onto an embedding space that does not have meaningful cluster geometry, producing near-random partitions that score close to 0 on the silhouette metric.

5.5 Method Agreement Analysis

5.5.1 HDBSCAN vs DEC Agreement on AST Embeddings

Table 5.5 presents the agreement metrics between HDBSCAN and DEC on each embedding space for id_00. Agreement is measured by the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), both computed over non-noise HDBSCAN-assigned samples (noise points labeled -1 by HDBSCAN are excluded from both sides of the comparison).

Table 5.5: HDBSCAN vs DEC agreement metrics per embedding space, pump id_00. ARI and NMI computed over non-noise HDBSCAN samples only.

Embedding	HDBSCAN k	DEC k	ARI	NMI	Interpretation
AST	3	4	0.2515	0.5047	Moderate agreement — shared structure, DEC subclusters one HDBSCAN group
CAE	2	1	0.0000	0.0000	No agreement — embedding space lacks cluster structure for both methods

The AST embedding space yields an ARI of 0.2515 and an NMI of 0.5047 between HDBSCAN and DEC. An ARI of 0.25 indicates that the two methods agree substantially beyond chance but do not produce identical partitions — expected, given that HDBSCAN finds 3 clusters and DEC finds 4. The NMI of 0.5047 indicates that knowing the assignment of one method reduces the uncertainty about the assignment of the other by approximately 50%, confirming a meaningful shared structure. The most likely explanation for the k discrepancy (3 vs 4) is that DEC is splitting one of the HDBSCAN clus-

ters into two sub-groups through its joint embedding-and-assignment optimization, which the confusion matrix heatmap can be used to verify.

The CAE embedding space yields $\text{ARI} = 0$ and $\text{NMI} = 0$, the two methods have nothing in common. This is the expected outcome when the underlying embedding space contains no cluster structure: HDBSCAN finds two tiny high-density pockets, while DEC is forced to assign all points to a single cluster ($k = 1$ in effect, despite receiving nominally k input from the generalizability criterion). The complete absence of agreement on CAE is itself a strong and informative negative result — it confirms that the cluster structure found in the AST space is not a generic property of the anomalous audio files themselves, but is genuinely tied to the richness of the AST representation.

5.5.2 Cross-Embedding Agreement

Beyond within-embedding method agreement, the cross-embedding ARI between DEC/AST and DEC/CAE is also computed. The resulting value of 0.2214 is low but non-zero. This value should be interpreted in the context of the full picture: DEC/AST finds 4 coherent clusters (Silhouette 0.39) while DEC/CAE effectively produces near-random assignments (Silhouette 0.06). The 0.22 cross-embedding ARI is therefore unlikely to reflect genuine shared cluster structure discovered by both embeddings; more plausibly, it reflects

mild correlation in the allocation of samples to groups driven by incidental properties of the data. This interpretation is consistent with the near-zero cross-embedding HDBSCAN ARI of 0.0680 (computed after excluding the 131 samples flagged as noise by either method), which provides the most conservative and reliable cross-embedding comparison.

5.6 Results Across All Machine IDs

5.6.1 Consolidated HDBSCAN Summary

The complete HDBSCAN analysis is repeated identically for pump machine IDs id_02, id_04, and id_06. Table 5.6 presents a consolidated summary of HDBSCAN results on AST embeddings across all four machine IDs.

Table 5.6: HDBSCAN clustering results on AST embeddings across all pump machine IDs (0 dB SNR). k^* (DEC) is the generalizability-selected cluster count. ARI is the HDBSCAN vs DEC agreement on AST, computed over non-noise samples.

Machine ID	N Anomalous	k (HDBSCAN)	Noise Frac.	Silhouette DBI		k^* (DEC)	ARI
id_00	143	3	0.035	0.4058	1.0266	4	0.2515
id_02	111	3	0.099	0.2952	1.1000	5	0.4608
id_04	100	2	0.160	0.2455	1.2330	5	0.1449
id_06	102	2	0.029	0.5221	0.7295	5	0.3271

Several consistent patterns emerge. First, the AST embedding space yields non-trivial cluster structure across all four machine IDs: HDBSCAN discovers 2–3 clusters in every case, with noise fractions

ranging from 2.9% (id_06) to 16.0% (id_04). This cross-machine consistency strongly supports the hypothesis that the AST embedding space reliably encodes fault-type variation, rather than capturing machine-specific idiosyncrasies. Second, the Silhouette Scores span a meaningful range: id_06 achieves the highest score (0.5221) with the lowest Davies-Bouldin Index (0.7295), indicating the cleanest cluster separation, while id_04 has the lowest Silhouette Score (0.2455) and the highest noise fraction, suggesting that id_04 fault acoustics are less clearly separated in the embedding space. Third, the generalizability criterion consistently selects $k^* = 5$ for DEC on id_02, id_04, and id_06, compared to $k^* = 4$ for id_00, suggesting that DEC identifies the finer substructure within HDBSCAN’s coarser density-based partitions across three of the four machines.

5.6.2 CAE Clustering Across Machine IDs

The CAE embedding results further reinforce the pattern observed for id_00: for id_04 and id_06, HDBSCAN assigns every single anomalous sample as noise (noise fraction = 1.000), finding no clusters whatsoever. For id_02, two clusters are nominally identified but the second cluster has a stability of 0.005 and the noise fraction remains at 29.7%, indicating that even the best CAE result across the four machines reflects only marginal structure. The contrast with the AST results, where every machine ID yields coherent clusters with meaningful silhouette scores, is unambiguous and consistent.

Table 5.7: HDBSCAN results on CAE embeddings across all pump machine IDs. A noise fraction of 1.000 indicates that HDBSCAN found no density-connected clusters whatsoever.

Machine ID	k (HDBSCAN)	Noise Frac.	Silhouette	DBI	Interpretation
id.00	2	0.916	0.3973	0.9266	Marginal structure in tiny non-noise fraction
id.02	2	0.297	0.1790	1.6984	Weak structure; second cluster near-degenerate (stability 0.005)
id.04	0	1.000	—	—	Complete noise; no clusters found
id.06	0	1.000	—	—	Complete noise; no clusters found

5.6.3 Per-Machine DEC Validity and Agreement

Table 5.8: DEC internal validity scores and HDBSCAN vs DEC agreement across all machine IDs. For id_04 and id_06, CAE ARI/NMI values of 1.0000 are trivially vacuous: both methods found zero CAE clusters (all samples assigned as noise), so there is no partition to compare.

Machine ID	DEC/AST Sil.	DEC/AST DBI	DEC/CAE Sil.	DEC/CAE DBI	AST ARI	AST NMI
id.00	0.3872	1.0325	0.0596	5.1139	0.2515	0.5047
id.02	0.2456	1.2672	0.0389	3.5474	0.4608	0.6476
id.04	0.1747	1.6766	0.0587	2.8689	0.1449	0.3658
id.06	0.2110	1.6802	0.0471	3.5567	0.3271	0.5643

Across all four machine IDs, the DEC Silhouette Score on CAE embeddings never exceeds 0.06, and the Davies-Bouldin Index never falls below 2.87, both indicating near-random partitions. In contrast, the DEC Silhouette Score in AST embeddings ranges from 0.17 to

0.39, and the HDBSCAN-DEC agreement on AST (measured by ARI) ranges from 0.14 to 0.46 with NMI values between 0.37 and 0.65. The highest agreement is observed for id_02 (ARI = 0.4608, NMI = 0.6476), indicating that the AST embedding space for id_02 anomalous samples has the strongest recoverable cluster structure of the four machines.

5.6.4 Spectral Characterisation Across Machine IDs

Spectral descriptors are computed for all machine IDs. Table 5.9 presents the per-cluster spectral centroid (the most discriminative descriptor) for id_02 and id_06, which produce the clearest HDBSCAN cluster structures in the AST embeddings after id_00.

Table 5.9: Per-cluster spectral centroid (Hz, mean $\pm$ std) from HDBSCAN on AST embeddings for id_02 and id_06. The consistent ordering across clusters within each machine indicates spectral separation driven by shape rather than amplitude differences (RMS is identical at 0.01 across all clusters and machines).

Machine ID	Clusters	Cluster 0 Centroid (Hz)	Cluster 1 Centroid (Hz)	Cluster 2 Centroid (Hz)
id_02	3	2295.92 $\pm$ 168.83	2159.96 $\pm$ 69.59	1523.17 $\pm$ 165.21
id_04	2	1814.25 $\pm$ 144.11	2089.42 $\pm$ 184.96	—
id_06	2	1606.01 $\pm$ 210.36	1170.14 $\pm$ 211.53	—

A notable cross-machine observation is that the absolute spectral centroid values differ substantially between machine IDs (id_02 clusters span $\approx$ 1500–2300 Hz, while id_06 clusters span $\approx$ 1170–1606 Hz). This is expected: each machine ID is a physically distinct pump unit with its own characteristic acoustic signature. The intra-machine cluster separation — the difference between centroids within

the same machine’s clusters — is the relevant measure of fault subtype discriminability. This separation ranges from approximately 300 Hz (id_02 Clusters 0 vs 1) to 436 Hz (id_06 Clusters 0 vs 1), indicating that the AST embedding consistently captures spectrally meaningful distinctions across machine units.

5.7 Discussion

- **Meaningful Structure in Anomalies:** The clustering results indicate that anomalous samples are not random but form coherent groups, suggesting underlying fault-specific patterns.
- **Consistency Across Methods:** Agreement between different clustering approaches demonstrates that the learned representations capture stable and reliable structures in the data.
- **Embedding Quality Matters:** Improved clustering performance in learned embedding spaces highlights the importance of high-quality feature representations for downstream analysis.
- **Variability Across Machine Units:** Differences in clustering behaviour across machine IDs indicate that anomaly characteristics are machine-specific and may require adaptive modeling.
- **Limitations in Ground Truth Validation:** The absence of explicit fault labels limits the ability to directly validate cluster-to-fault correspondence, making interpretation partially inferential.

Conclusion and Future Work

This chapter summarizes the key findings of the study and reflects on the overall contributions of the proposed framework. It also outlines the limitations of the current work and highlights potential directions for future improvements and extensions [33, 12, 25].

6.1 Conclusion

This work presented an acoustic-based framework for industrial anomaly detection, focusing on both detection performance and post-detection analysis. The baseline system demonstrated the limitations of reconstruction-based approaches, particularly in handling complex acoustic variations and subtle fault patterns.

To address these limitations, a representation-driven approach was introduced, combining learned audio embeddings with a one-class classification framework. This approach significantly improved anomaly detection performance, achieving a substantial gain over conventional methods and providing more stable and consistent results across different machine conditions.

Beyond detection, the study further explored the structure of anomalous data using clustering techniques. The results showed that anomalies are not random but exhibit meaningful groupings, indicating the presence of underlying fault patterns. This highlights the importance of moving beyond binary detection toward more informative analysis.

Overall, the proposed framework demonstrates that combining representation learning, anomaly detection, and clustering can provide a more comprehensive solution for intelligent machine monitoring and predictive maintenance [34, 35, 36].

6.2 Future Work

- **Evaluation Under Diverse Conditions:** Extending the evaluation to multiple noise levels and varying environmental conditions would provide a more comprehensive assessment of system robustness.
- **Improving Training Stability:** Further work is required to stabilise the optimisation process, particularly for the one-class classification stage, to ensure consistent performance.
- **Cross-Domain Generalisation:** Adapting the framework to different machine types and industrial settings would improve its practical applicability.
- **Integration into Real-Time Systems:** Developing a real-

time deployment pipeline would enable continuous monitoring and faster response to anomalies.

- **Incorporating Ground Truth Validation:** Introducing labeled fault data would allow more rigorous validation of clustering results and improve interpretability.

Bibliography

- [1] Rohan Giri et al. “Self-Supervised Learning for Anomalous Sound Detection”. In: *DCASE Workshop*. 2020.
- [2] Harsh Purohit et al. “MIMII Dataset: Sound Dataset for Malfunctioning Industrial Machine Investigation and Inspection”. In: *Proceedings of the Detection and Classification of Acoustic Scenes and Events (DCASE)*. 2019, pp. 209–213.
- [3] Yuma Koizumi et al. “DCASE 2020 Challenge Task 2: Unsupervised Detection of Anomalous Sounds for Machine Condition Monitoring”. In: *DCASE Challenge*. 2020.
- [4] Varun Chandola, Arindam Banerjee, and Vipin Kumar. “Anomaly Detection: A Survey”. In: *ACM Computing Surveys* (2009).
- [5] Yuma Koizumi et al. “Unsupervised Detection of Anomalous Sound Using Deep Learning”. In: *DCASE Workshop*. 2017.
- [6] Yuan Gong, Yu-An Chung, and James Glass. “AST: Audio Spectrogram Transformer”. In: *Proceedings of Interspeech*. 2021.
- [7] Erik Marchi et al. “A Novel Approach for Automatic Acoustic Anomaly Detection Using Autoencoders”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2015.
- [8] Yuma Koizumi et al. “Unsupervised Detection of Anomalous Sound Based on Deep Learning and the Neyman–Pearson Lemma”. In: *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 27.1 (2019), pp. 212–224. DOI: [10.1109/TASLP.2018.2877258](https://doi.org/10.1109/TASLP.2018.2877258).
- [9] Guansong Pang et al. “Deep Learning for Anomaly Detection: A Review”. In: *ACM Computing Surveys* (2021).
- [10] Lukas Ruff et al. “A Unifying Review of Deep and Shallow Anomaly Detection”. In: *Proceedings of the IEEE* (2021).
- [11] Ryo Tanabe et al. “MIMII-DUE: Dataset for Unsupervised Anomaly Detection of Industrial Machine Sounds”. In: *Proceedings of the Detection and Classification of Acoustic Scenes and Events (DCASE)*. 2022.
- [12] Erik Vanem and Andreas Brandsæter. “Unsupervised anomaly detection based on clustering methods and sensor data on a marine diesel engine”. In: *Journal of Marine Engineering & Technology* 20.4 (2021), pp. 217–234. DOI: [10.1080/20464177.2019.1633223](https://doi.org/10.1080/20464177.2019.1633223).
- [13] Ricardo Campello, Davoud Moulavi, and Jörg Sander. “Density-Based Clustering Based on Hierarchical Density Estimates”. In: *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)* (2013).
- [14] Junyuan Xie, Ross Girshick, and Ali Farhadi. “Unsupervised Deep Embedding for Clustering Analysis”. In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2016.

- [15] Geoffrey E. Hinton and Ruslan R. Salakhutdinov. “Reducing the Dimensionality of Data with Neural Networks”. In: *Science* 313.5786 (2006), pp. 504–507. DOI: [10.1126/science.1127647](https://doi.org/10.1126/science.1127647).
- [16] Mayu Sakurada and Takehisa Yairi. “Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction”. In: *Proceedings of the ACM Workshop on Machine Learning for Sensory Data Analysis (MLSDA)*. 2014, pp. 4–11.
- [17] W. J. Youden. “Index for rating diagnostic tests”. In: *Cancer* (1950).
- [18] Alan V. Oppenheim and Alan S. Willsky. *Signals and Systems*. Prentice Hall, 1999.
- [19] Brian McFee et al. “librosa: Audio and Music Signal Analysis in Python”. In: *Proceedings of the Python in Science Conference (SciPy)*. 2015.
- [20] Shawn Hershey et al. “CNN Architectures for Large-Scale Audio Classification”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2017.
- [21] Sergey Ioffe and Christian Szegedy. “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift”. In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2015.
- [22] Diederik Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *International Conference on Learning Representations (ICLR)*. 2015.
- [23] Tom Fawcett. “An Introduction to ROC Analysis”. In: *Pattern Recognition Letters* (2006).
- [24] Jort F. Gemmeke et al. “Audio Set: An ontology and human-labeled dataset for audio events”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2017.
- [25] Ashish Vaswani et al. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017.
- [26] Alexey Dosovitskiy et al. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”. In: *International Conference on Learning Representations (ICLR)*. 2021.
- [27] Lukas Ruff et al. “Deep One-Class Classification”. In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2018.
- [28] Bernhard Schölkopf et al. “Estimating the Support of a High-Dimensional Distribution”. In: *Neural Computation* (2001).
- [29] John Healy and Leland McInnes. “Uniform Manifold Approximation and Projection”. In: *Nature Reviews Methods Primers* (2024).
- [30] Martin Ester et al. “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases”. In: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 1996, pp. 226–231.

- [31] Peter J. Rousseeuw. “Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis”. In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65.
- [32] Takaya Saito and Marc Rehmsmeier. “The Precision-Recall Plot is More Informative than the ROC Plot”. In: *PLOS ONE* (2015).
- [33] Markus Breunig et al. “LOF: Identifying Density-Based Local Outliers”. In: *Proceedings of the ACM SIGMOD* (2000).
- [34] Ilya Loshchilov and Frank Hutter. “Decoupled Weight Decay Regularization”. In: *International Conference on Learning Representations (ICLR)* (2019).
- [35] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. “Isolation Forest”. In: *Proceedings of the IEEE International Conference on Data Mining (ICDM)*. 2008, pp. 413–422.
- [36] Peter Rousseeuw. “Silhouettes: A Graphical Aid to the Interpretation of Cluster Analysis”. In: *Journal of Computational and Applied Mathematics* (1987).